The Primordial Lithium Problem and FFGFT

A Brief Summary for the IPI Correspondence

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GitHub: https://github.com/jpascher/T0-Time-Mass-Duality

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Abstract

The primordial lithium problem – the persistent factor 3–4 discrepancy ($4{,}2\sigma$–$5{,}3\sigma$) between Big Bang Nucleosynthesis (BBN) predictions for ^{7}Li and observations of metal-poor halo stars (the Spite plateau) – has resisted resolution for four decades. Standard solution classes (astrophysical depletion, nuclear physics, beyond-Standard-Model physics) all face structural difficulties. Fields (2011) explicitly lists "large changes to the cosmological framework" as a possible solution class. FFGFT instantiates that class: in a static, eternal ξ-universe there is no primordial nucleosynthesis epoch, and the Spite plateau is not the relic of an expanding-universe event. This note presents the FFGFT position, including a tentative explanation sketch for why D and ^{4}He abundances nonetheless agree with BBN+CMB predictions while ^{7}Li does not. The explanation is presented as a sketch, not a closed derivation.

1 The Problem in Numbers

The Big Bang Nucleosynthesis prediction, when fed the WMAP/Planck baryon-to-photon ratio $\eta_{10} = 6{,}23 \pm 0{,}17$, yields:

$$\text{D/H}\big|_{\text{BBN+CMB}} = (2{,}82 \pm 0{,}21) \times 10^{-5} \qquad \text{(observation: matches)} \qquad (1)$$

$$Y_p(^4\text{He}) = 0{,}249 \pm 0{,}009 \qquad \text{(observation: matches)} \qquad (2)$$

$$^7\text{Li/H}\big|_{\text{BBN+CMB}} = (5{,}24^{+0{,}71}_{-0{,}67}) \times 10^{-10} \qquad \text{(observation: differs)} \qquad (3)$$

The Spite-plateau measurement [1, 6] of metal-poor halo stars gives

$$^7\text{Li/H}\big|_{\text{obs}} = (1{,}23^{+0{,}68}_{-0{,}32}) \times 10^{-10}\,, \qquad (4)$$

a factor of $2{,}4$–$4{,}3$ below the BBN+CMB prediction, corresponding to $4{,}2\sigma$–$5{,}3\sigma$ [4]. Cyburt, Fields and Olive titled their 2008 update "A Bitter Pill: The Primordial Lithium Problem Worsens" – improved nuclear cross-section data *increased* the predicted ^{7}Li abundance.

The discrepancy is universal across stellar environments: the Spite plateau has been observed in halo field stars, globular clusters [4], dwarf spheroidals, ultra-faint galaxies, and accreted satellites [9]. It is not a feature of any one galactic environment.

2 Standard Solution Classes

Following Fields (2011), three classes have been pursued:

Astrophysical (stellar depletion). The most active current programme [9] models internal stellar mixing – atomic diffusion, parametric turbulence, rotation-induced angular momentum transport – to deplete photospheric ^{7}Li by a factor ~ 3. The challenge: the Spite plateau is observationally *thin*, with little star-to-star scatter, and is found across very different stellar populations. Universal depletion across these environments is difficult to motivate physically.

Nuclear physics. Hypothetical resonances enhancing ^{7}Be destruction, e.g. ^{7}Be+^{3}He $\rightarrow$ ^{10}C* $\rightarrow$ $p + {}^9$B [6, 7], or revised ^{3}He$(\alpha, \gamma)^7$Be cross-sections. Recent experimental work has *tightened* the rates and *worsened* the prediction [4].

Beyond Standard Model. Decaying SUSY particles, time-varying fundamental constants [8], primordial magnetic fields. Constraint: D and ^{4}He must remain unperturbed, which leaves a narrow parameter window.

The fourth class. Fields (2011) closes his classification with: *"large changes to the cosmological framework used to interpret light element data."* This category has received less attention but is logically open.

3 The FFGFT Position

FFGFT instantiates the fourth class. The cosmological framework is static and eternal, derived from the single dimensionless parameter $\xi = 4/30{,}000$:

- No Big Bang. The Heisenberg time-energy uncertainty argument (Doc 025) precludes a finite-duration cosmological singularity.
- No expanding spacetime. Cosmological redshift is a geometric path effect through the granular ξ-vacuum, not a Doppler signature (Doc 026).
- No primordial nucleosynthesis epoch. The Spite plateau is therefore *not* the relic of a universe-wide event 13.8 Gyr ago.

In this framework, the Spite plateau records the long-term equilibrium ^{7}Li abundance produced and maintained by stellar processes (Galactic cosmic-ray spallation, novae, AGB stars, core-collapse supernovae, white-dwarf mergers [9]) in a quasi-stationary state. The factor 3–4 "discrepancy" is then not a discrepancy with reality, but with a prediction that requires an event (BBN) that did not occur in this framework.

The honest follow-up question is: *why do D and ^{4}He match BBN+CMB so precisely if there is no BBN?*

4 Explanation Sketch: Why D and ^{4}He Match

What follows is an *explanation sketch*, not a closed derivation. The point is to indicate a plausible mechanism, not to claim that the mechanism is already worked out.

Step 1 – The CMB is real, but not a relic. FFGFT accepts the CMB as a physical phenomenon, but identifies it as a manifestation of the ξ-vacuum rather than the redshifted afterglow of a hot early phase. In Doc 091, the CMB energy density is fixed by

$$\rho_{\mathsf{CMB}} = \frac{\xi\,\hbar c}{L_\xi^4}\,, \qquad L_\xi \approx 100{,}24\,\mu\mathrm{m}\,, \tag{5}$$

and the CMB temperature follows accordingly. The photon bath that carries the spectrum is real and operative *now*, not just at $z \approx 1100$.

Step 2 – The baryon-to-photon ratio. In standard BBN, light-element yields depend on the single parameter $\eta = n_B/n_\gamma$. In FFGFT, both the photon density (from ρ_{CMB} and T_{CMB}) and the cosmic baryon density (from continuous stellar processes integrated over an eternal universe) are present-day quantities. Their ratio defines an *effective* η_{eff}, which the WMAP/Planck inference correctly extracts from the CMB acoustic peaks – because the acoustic-peak structure depends only on the present-day η, not on its history.

The ΛCDM interpretation reads back from this η to a primordial η via assumed entropy conservation; FFGFT keeps the present-day η as a stationary characteristic of the ξ-universe.

Step 3 – Thermal-equilibrium yields. For light nuclides whose abundances are governed by *rapid two-body equilibrium reactions in a thermal bath*, the equilibrium abundance is a function of η and T alone, not of the history that produced this (η, T) pair. The reaction network that determines D and ^{4}He abundances under BBN conditions is dominated by such two-body equilibria with strong reverse reactions:

$$p + n \leftrightarrow d + \gamma, \qquad d + d \leftrightarrow {}^3\mathrm{He} + n, \qquad {}^3\mathrm{He} + d \leftrightarrow {}^4\mathrm{He} + p\,, \ldots \tag{6}$$

If the FFGFT ξ-vacuum sustains a $(T_{\mathsf{CMB}}, \eta_{\mathsf{eff}})$ that coincides with the WMAP-inferred values, the same equilibrium abundances follow. This is why D and ^{4}He observations agree with BBN+CMB predictions: the predictions describe a *thermal equilibrium*, and the equilibrium is realised in FFGFT for the same (T, η) by construction. The agreement is not a coincidence; it is a consistency check on the ξ-determination of T_{CMB}.

Step 4 – Why ^{7}Li breaks the pattern. ^{7}Li is the exception because its production-destruction chain has a *slow, non-equilibrium* component:

1. ^{7}Li at WMAP η is produced predominantly as ^{7}Be via ^{3}He$(\alpha, \gamma)^7$Be [6].
2. In standard BBN, ^{7}Be then survives the cessation of nuclear reactions and decays to ^{7}Li by electron capture on a timescale of ~ 53 days – after BBN has frozen out.

3. In FFGFT, no freeze-out occurs because there is no expansion-driven temperature drop. The ^{7}Be population sits in a stationary environment with continuous photon and electron interactions, including channels that destroy ^{7}Be on timescales much shorter than its laboratory half-life would suggest.

The BBN calculation assumes a clean ^{7}Be $\to$ ^{7}Li conversion after freeze-out, treating it as bookkeeping. In FFGFT this assumption is invalid: the stationary state allows continuous ^{7}Be destruction (via cosmic-ray-induced spallation, energetic photon channels in the ξ-vacuum, or stellar incorporation) that the BBN bookkeeping does not account for. The plateau-level ^{7}Li abundance is then the stationary attractor of the full network in a static universe, and it is naturally lower than the BBN bookkeeping value.

What this sketch does and does not claim. *Does claim:* a structural reason why an equilibrium-dominated nuclide (D, ^{4}He) follows the standard prediction while a freeze-out-dependent nuclide (^{7}Li) does not. *Does not claim:* a closed quantitative derivation of the Spite-plateau value from ξ. The next step would be to set up the stationary network in the ξ-vacuum and compute the attractor for ^{7}Li. This is open work, not a finished result.

5 Summary

1. The lithium problem is real and standard: factor 3–4, $4{,}2\sigma$–$5{,}3\sigma$, persistent across four decades and many proposed fixes.
2. Astrophysical, nuclear and BSM solutions all face structural challenges; the cosmological-framework alternative (Fields 2011) remains open.
3. FFGFT instantiates the cosmological-framework alternative: no Big Bang, no primordial nucleosynthesis epoch, the Spite plateau is a stationary stellar-equilibrium feature.
4. The agreement of D and ^{4}He with BBN+CMB is not coincidental but a consistency check: equilibrium abundances depend on (T, η) alone, and FFGFT fixes both via ξ-geometry.
5. ^{7}Li differs because its abundance depends on a freeze-out assumption that is invalid in a static universe; the stationary attractor lies below the BBN bookkeeping value.
6. The quantitative derivation of the FFGFT ^{7}Li attractor is open work; this note presents the explanation as a sketch.

References

References

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FFGFT source documents: Doc 025 (T0 Cosmology, ξ-field universe and CMB), Doc 026 (geometric cosmology, H_0 and the redshift mechanism), Doc 091 (Casimir-CMB relation, L_ξ). All available at `https://github.com/jpascher/T0-Time-Mass-Duality` and Zenodo v1.0.13: `https://zenodo.org/records/20041529`.